Bayesian Optimization Report: Function 7

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.706	Good
2	0.705	Good
3	0.705	Good
4	0.704	Good
5	0.703	Good
6	0.696	Medium
7	0.697	Medium
8	0.699	Medium
9	0.697	Medium
10	0.696	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.0910	0.0187	0.1962
2	0.0813	0.0258	0.2380
3	0.0664	0.0165	0.2260
4	0.0555	0.0167	0.1523
5	0.0506	0.0163	0.2014
6	0.0466	0.0166	0.1672
7	0.0464	0.0141	0.1516
8	0.0438	0.0152	0.1653
9	0.0434	0.0137	0.1360
10	0.0426	0.0132	0.1474

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies

1	[0.04847286 0.02135972 0.22247663 0.27519554 0.29470816 0.63684871]	1.6870018491721683	Matern(ν=0.5)	PORTFOLIO	0.706	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(0.6494487743469476)}}
2	[0. 0. 0.32779432 0.29980551 0.42089648 0.59156801]	1.447175885389588	Matern(ν=0.5)	PORTFOLIO	0.705	None
3	[0. 0. 0.22609121 0.15793527 0.27994077 0.56449273]	1.755230730320253	Matern(ν=0.5)	PORTFOLIO	0.705	None
4	[0.09605551 0. 0.2560619 0.20344109 0.36668138 0.56678936]	1.9349636830135686	Matern(ν=0.5)	PORTFOLIO	0.704	None
5	[0.01005668 0. 0.23848072 0.1946851 0.33263739 0.61714902]	2.004900575163112	Matern(ν=0.5)	PORTFOLIO	0.703	None
6	[0.03134649 0.00757786 0.38217671 0.22222354 0.33987712 0.58543274]	1.9898209623015406	Matern(ν=0.5)	PORTFOLIO	0.696	None
7	[0. 0.02559569 0.33650014 0.2076399 0.32827593 0.59742904]	2.016129309622693	Matern(ν=0.5)	PORTFOLIO	0.697	None
8	[0.020928 0.033734 0.21098883 0.21274554 0.33901628 0.58253575]	1.998528421567837	Matern(ν=0.5)	PORTFOLIO	0.699	None
9	[0. 0. 0.30029238 0.19602686 0.35115417 0.59046292]	2.0255743469364145	Matern(ν=0.5)	PORTFOLIO	0.697	None
10	[0. 0. 0.36122567 0.21512305 0.35870951 0.60377551]	2.0034219969822247	Matern(ν=0.5)	PORTFOLIO	0.696	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.588432
EI	0.009070
PI	0.309998
THOMPSON	1.542545
KG	0.009070
ENTROPY	2.528925
PORTFOLIO	1.250993

Iteration 2

Method	Max Value
UCB	1.451543
EI	0.004915
PI	0.205921
THOMPSON	1.463465
KG	0.004915
ENTROPY	2.446931
PORTFOLIO	1.186779

Iteration 3

Method	Max Value
UCB	1.447083
EI	0.003998
PI	0.279631
THOMPSON	1.461566
KG	0.003998
ENTROPY	2.368539
PORTFOLIO	1.105440

Iteration 4

Method	Max Value
UCB	1.445413
EI	0.003878
PI	0.244803
THOMPSON	1.441649
KG	0.003878
ENTROPY	2.308687
PORTFOLIO	1.049602

Iteration 5

Method	Max Value
UCB	1.441679
EI	0.003680

PI	0.213718
THOMPSON	1.445287
KG	0.003680
ENTROPY	2.237469
PORTFOLIO	1.017285

Iteration 6

Method	Max Value
UCB	1.425838
EI	0.003023
PI	0.225977
THOMPSON	1.453691
KG	0.003023
ENTROPY	2.202869
PORTFOLIO	0.990380

Iteration 7

Method	Max Value
UCB	1.427956
EI	0.002838
PI	0.208592
THOMPSON	1.432340
KG	0.002838
ENTROPY	2.174543
PORTFOLIO	0.986471

Iteration 8

Method	Max Value
UCB	1.428318
EI	0.002094
PI	0.214798
THOMPSON	1.428760
KG	0.002094
ENTROPY	2.174830
PORTFOLIO	0.976628

Iteration 9

Method	Max Value
UCB	1.425535
EI	0.002418
PI	0.224741
THOMPSON	1.435849
KG	0.002418
ENTROPY	2.160423
PORTFOLIO	0.969194

Iteration 10

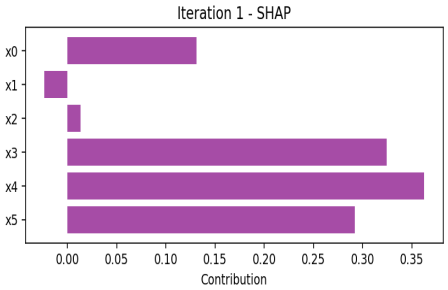
Method	Max Value
UCB	1.423230
EI	0.001709
PI	0.205681
THOMPSON	1.419485
KG	0.001709
ENTROPY	2.156830
PORTFOLIO	0.968100

LOO MSE Summary

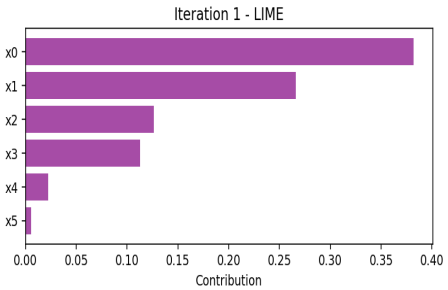
Iteration	LOO MSE
1	0.049595
2	0.046918
3	0.037289
4	0.029602
5	0.026955
6	0.022457
7	0.021509
8	0.020511
9	0.020063
10	0.019492

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



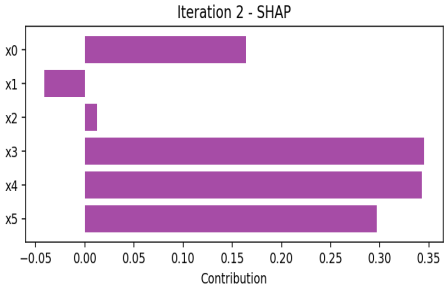
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.13136
x1	-0.02339
x2	0.01335
x3	0.32476
x4	0.36280
x5	0.29234

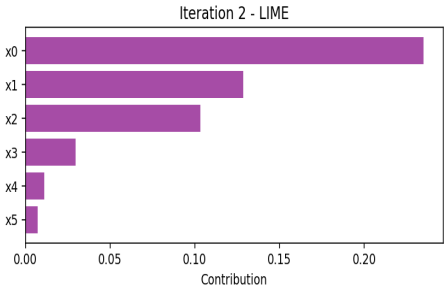
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.38178
x1	0.26605
x2	0.12630
x3	0.11300
x4	0.02246
x5	0.00541

Iteration 2 - SHAP



Iteration 2 - LIME



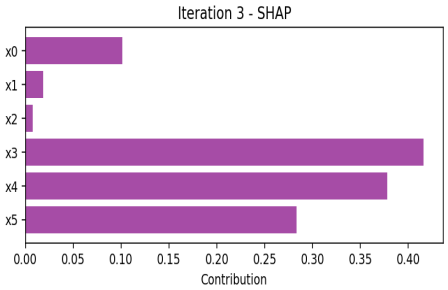
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.16394
x1	-0.04122
x2	0.01256
x3	0.34536
x4	0.34296
x5	0.29712

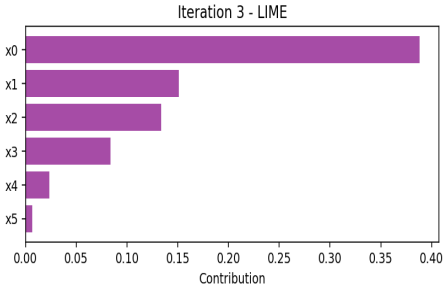
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.23504
x1	0.12863
x2	0.10328
x3	0.02962
x4	0.01119
x5	0.00742

Iteration 3 - SHAP



Iteration 3 - LIME



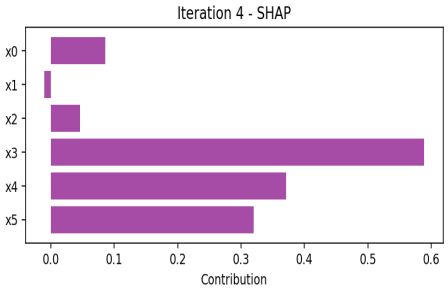
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.10135
x1	0.01842
x2	0.00742
x3	0.41579
x4	0.37825
x5	0.28319

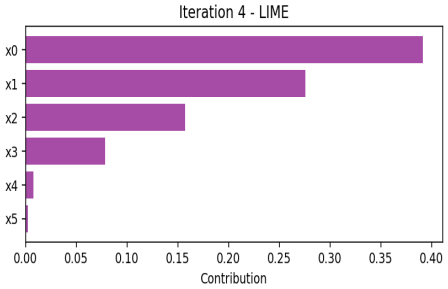
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.38802
x1	0.15102
x2	0.13390
x3	0.08354
x4	0.02361
x5	0.00656

Iteration 4 - SHAP



Iteration 4 - LIME



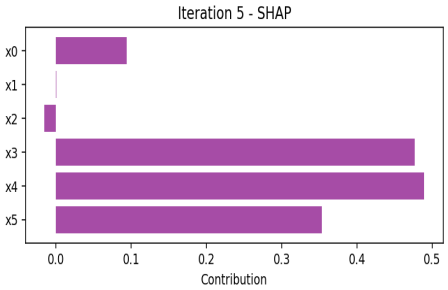
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.08548
x1	-0.01027
x2	0.04581
x3	0.58895
x4	0.37067
x5	0.32023

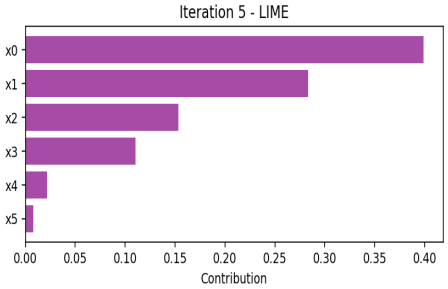
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.39130
x1	0.27581
x2	0.15742
x3	0.07839
x4	0.00775
x5	0.00229

Iteration 5 - SHAP



Iteration 5 - LIME



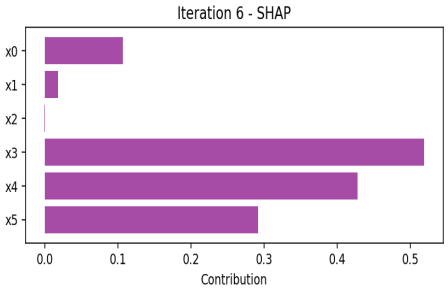
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.09443
x1	0.00093
x2	-0.01522
x3	0.47687
x4	0.48966
x5	0.35381

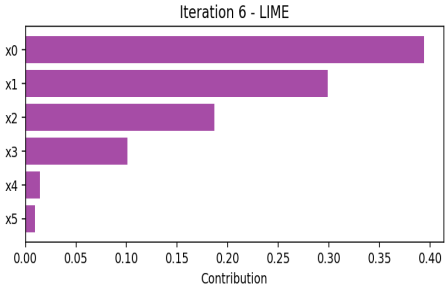
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.39887
x1	0.28345
x2	0.15312
x3	0.11063
x4	0.02167
x5	0.00804

Iteration 6 - SHAP



Iteration 6 - LIME



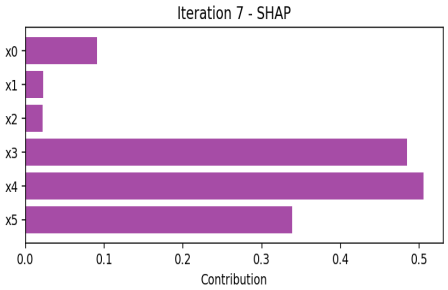
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.10629
x1	0.01778
x2	-0.00094
x3	0.51906
x4	0.42737
x5	0.29146

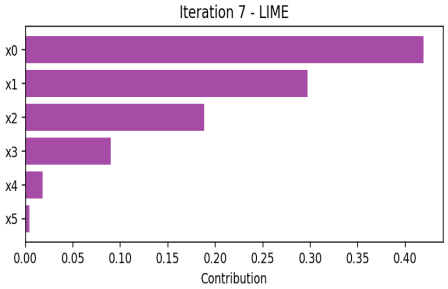
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.39381
x1	0.29896
x2	0.18658
x3	0.10081
x4	0.01456
x5	0.00921

Iteration 7 - SHAP



Iteration 7 - LIME



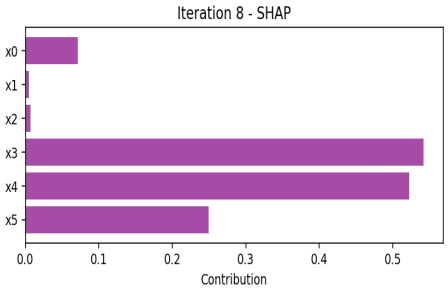
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.09122
x1	0.02243
x2	0.02169
x3	0.48446
x4	0.50518
x5	0.33892

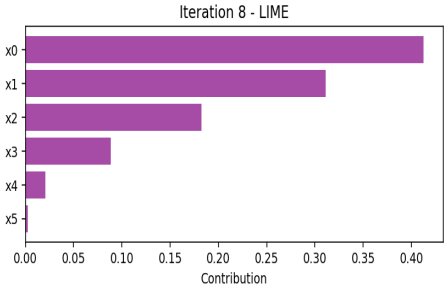
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.41919
x1	0.29708
x2	0.18808
x3	0.09013
x4	0.01827
x5	0.00409

Iteration 8 - SHAP



Iteration 8 - LIME



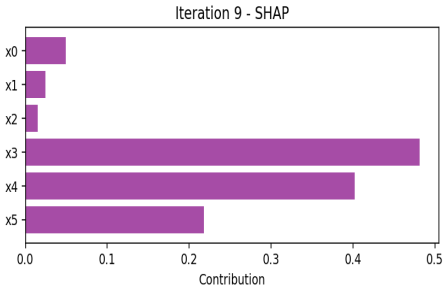
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.07151
x1	0.00512
x2	0.00691
x3	0.54193
x4	0.52238
x5	0.24973

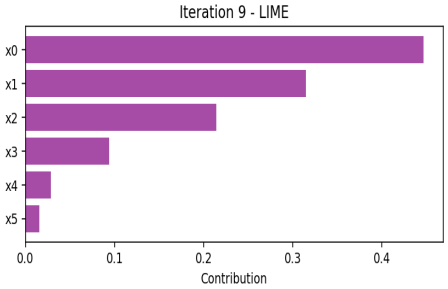
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.41249
x1	0.31124
x2	0.18236
x3	0.08867
x4	0.02092
x5	0.00248

Iteration 9 - SHAP



Iteration 9 - LIME



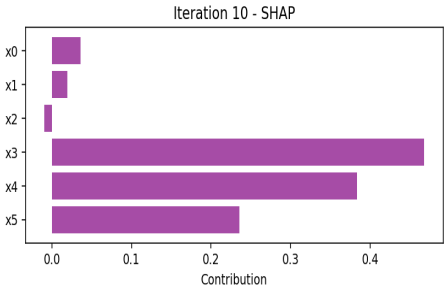
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.04912
x1	0.02467
x2	0.01486
x3	0.48120
x4	0.40198
x5	0.21825

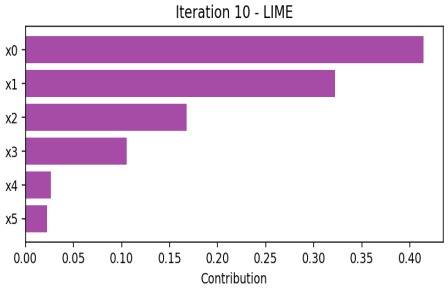
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.44657
x1	0.31467
x2	0.21395
x3	0.09413
x4	0.02871
x5	0.01571

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.03626
x1	0.01983
x2	-0.00936
x3	0.46809
x4	0.38378
x5	0.23586

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.41403
x1	0.32191
x2	0.16789
x3	0.10522
x4	0.02663
x5	0.02235